

TIANER ZHOU

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🎓 EDUCATION

University of Michigan, Ann Arbor, USA

2019 – 2022

Bachelor of Science in Computer Science (Summa Cum Laude)

- **GPA:** 3.917/4.0
- **Core Courses:** Operating Systems, Computer Architecture, Database, Computer Security, Data Structures & Algorithms, Linear Algebra, Applied Linear Regression, Machine Learning, Web Systems, Distributed Systems.
- **Practical Projects:**
 - Operating system kernel development (thread library, virtual memory pager, file system).
 - Distributed system implementation (Paxos architecture, sharded worker).
 - Arduino drone control system based on PID algorithm (Top 5% in class).

👤 WORK EXPERIENCE

Huawei Technologies Co., Ltd., Hangzhou

Aug 2022 – Present

Graphics Engineer HarmonyOS Graphics Foundation, now deployed on 50M+ devices.

- **Graphics Systems & Shader Optimization:** Developed 2D visual-effect shaders in the HarmonyOS graphics pipeline. Designed and applied automated parameter search algorithms (randomized genetic algorithms) to optimize shader parameters, improving runtime performance up to 10% in particular shaders and significantly reducing manual tuning.
- **Learning-based Rendering Exploration:** Participated in the development of native 3D Gaussian rendering; explored model-based 3D generation in resource-constrained mobile devices.

UI Framework Engineer native UI framework of HarmonyOS, supporting the development of 20,000+ apps.

- **Open Source Community Leader:** Core contributor to the UI engine repository (C++/TypeScript), ranked Top 1 with 1200+ commits among 1300+ contributors.[\[repo link\]](#)
- **Performance Optimization Expert:**
 - Built performant scroll containers (lazy loading, reuse, preloading), cut jank rate by 60%.
 - Designed image loader (deduplication, caching), multi-image speed matches Glide.
 - Created ReactNative-SVG renderer for HarmonyOS, 20% faster than Android solution.

🏠 PERSONAL PROJECTS

- **GPT Implementation:** Built modules including BPE tokenizer, transformer architecture, distributed KV cache, RoPE, flash attention kernel, post-training gRPO, etc. [\[repo link\]](#)
- **Reinforcement Learning Framework:** Implemented SAC/IQL/AWAC/PPO algorithms; IQL achieved 230/300 score on Ant Maze task (30,000 training steps)[\[repo link\]](#)
- **LLM Open Source Contributions:** Merged multiple PRs to vLLM and Nvidia Dynamo projects.
- **Agentic:** Developed HarmonyOS MCP, enabling LLMs to interact with HarmonyOS device UI.
- **Compiler Development:** Developed x86 compiler for JavaScript syntax in Rust: supports dynamic type conversion, closures with implicit variable capture [\[repo link\]](#)
- **OS Kernel Development:** Implemented multiple modules for xv6 (Unix-like kernel): pipe, fork, mmap, virtual memory pager, kernel trap, NIC driver.

⚙️ TECHNICAL SKILLS

Programming Languages: C++, TypeScript, Rust, Go, Python · **UI Frameworks:** SwiftUI, React · **System Design:** Distributed Systems, Observability, API Design · **Toolchain:** CI, Ninja, GN, gTest, gMock